

MISC

Quadratic formula:

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Monotonically in-/decreasing:

$$\forall x. (f'(x) > 0) \Rightarrow a > b \Rightarrow f(a) > f(b)$$
$$\forall x. (f'(x) < 0) \Rightarrow a > b \Rightarrow f(a) < f(b)$$

TRIGONOMETRY

x	0	$30^\circ = \pi/6$	$45^\circ = \pi/4$	$60^\circ = \pi/3$	$90^\circ = \pi/2$
$\sin(x)$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos(x)$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan(x)$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	

$\tan(x) = \frac{\sin(x)}{\cos(x)}$
$$\sin(2x) = 2 \sin(x) \cos(x)$$
$$\sin^2(x) + \cos^2(x) = 1$$
$$\mathbf{1, \sin(\alpha), \cos(\alpha)}$$

VECTORS

Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v^T v}$ $\nabla \ v\ = \frac{1}{\ v\ } v^T$
Distance	$d(u, v) = \ u - v\ $ $\nabla d(u, v) = \frac{1}{d(u, v)} ((u - v)^T (v - u)^T)$
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$ $\nabla (u \bullet v) = (v^T \ u^T)$

MATRICES

Determinant:

$$\det(a) = a$$
$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$
$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

$$\det(A \cdot B) = \det(A) \cdot \det(B) \quad \det(\mathbb{1}) = 1$$
$$\det(A^{-1}) = \frac{1}{\det(A)} \quad \det(A^T) = \det(A)$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^T)^T = A \quad (A+B)^T = A^T + B^T$$
$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5) \quad (\lambda A)^T = \lambda A^T \quad (AB)^T = B^T A^T$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$
$$= \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

Affine transformation:

$$M \in \mathbb{R}^{m \times n}, \quad A_{b,M} : \begin{cases} \mathbb{R}^m \rightarrow \mathbb{R}^n \\ x \mapsto b + M \cdot x \end{cases}$$

PROPERTIES

$$A \cdot A^{-1} = \mathbb{1} \quad \ker A = \{v \mid Av = 0\}$$
$$A + B = B + A \quad A \cdot B \neq B \cdot A$$
$$A + (B + C) = (A + B) + C \quad A(BC) = (AB)C$$
$$A(B + C) = AB + AC \quad (A + B)C = AC + BC$$

INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, \quad A \cdot B \in \mathbb{R}^{n \times r}$$
$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$
$$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ki}$$
$$(A \cdot B^T)_{ij} = \sum_k A_{ik} B_{kj}^T = \sum_k A_{ik} B_{jk}$$
$$a^T \cdot B \cdot b = \sum_{ij} (a^T)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$
$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

ADDITION/SUBTRACTION

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{n \times m}$$
$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(A - B)_{ij} = A_{ij} - B_{ij}$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$$
$$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M \cdot x$$
$$L_A(\vec{0}) = \vec{0}$$

DERIVATIVES

1	$\rightarrow 0$	x	$\rightarrow 1$
x^2	$\rightarrow 2x$	x^n	$\rightarrow nx^{n-1}$
$\frac{1}{x}$	$\rightarrow -\frac{1}{x^2}$	$\sqrt{x}$	$\rightarrow \frac{1}{2\sqrt{x}}$
e^x	$\rightarrow e^x$	e^{-x}	$\rightarrow -e^{-x}$
$\ln(x)$	$\rightarrow \frac{1}{x}$ for $x > 0$	$\ln(y \cdot x)$	$\rightarrow \frac{1}{x}$ for $x > 0$
a^x	$\rightarrow \ln(a) \cdot a^x$ for $a > 0, a \neq 1$	$\log_a(x)$	$\rightarrow \frac{1}{\ln(b) \cdot x}$
$\sin(x)$	$\rightarrow \cos(x)$	$\sin(2x)$	$\rightarrow 2 \cos(2x)$
$\cos(x)$	$\rightarrow -\sin(x)$	$\cos(ax)$	$\rightarrow -a \sin(ax)$
$\tan(x)$	$\rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\operatorname{arccot}(x)$	$\rightarrow -\frac{1}{1+x^2}$
$\cot(x)$	$\rightarrow -\frac{1}{\sin^2(x)}$	$\arcsin(x)$	$\rightarrow \frac{1}{\sqrt{1-x^2}}$
$\arccos(x)$	$\rightarrow -\frac{1}{\sqrt{1-x^2}}$	$\arctan(x)$	$\rightarrow \frac{1}{1+x^2}$

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Term	Definition
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha \cdot f(x))' = \alpha \cdot f'(x)$
Product rule	$(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES

Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
Softmax function	TODO:
RSS	$RSS = \sum_i (y_i - f(x_i))^2$
n -dimensional curve	$c: \mathbb{R} \rightarrow \mathbb{R}^n$
n -dimensional hyper-surface	$s: \mathbb{R}^n \rightarrow \mathbb{R}$
Reparametrization	$c(t) \rightarrow d(s) = c(h(s))$

Sigmoid function:

$$\sigma(x) : \mathbb{R} \rightarrow [0; 1] = \frac{1}{1 + e^{-x}}$$

DERIVATIVES

Function	Derivative
$f: \begin{cases} \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto y = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \frac{d}{dx} f_2(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix} \end{cases}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} = \nabla f \\ x \mapsto y = \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right) \end{cases}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^m \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0} \\ x \mapsto y = \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix} \end{cases}$

PARTIAL VS TOTAL DERIVATIVE

TODO:

example FS2024

$$f(x, y) = x^2 + 2y$$
$$\frac{\partial f}{\partial x} = 2x$$
$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 2x + 2 \cdot \frac{dy}{dx} \quad (y \text{ dependent on } x)$$

GRADIENTS

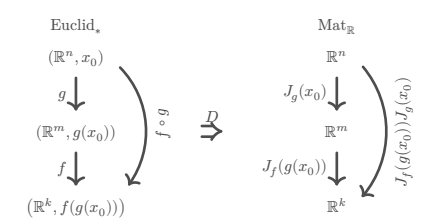
Common gradients:

$$f(x) = \|x\|^2 \quad \Rightarrow \nabla f = 2x$$
$$f(x) = a^T x + c \quad \Rightarrow \nabla f = a$$
$$f(x) = x^T A \quad \Rightarrow \nabla f = (A + A^T)x$$
$$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow \nabla f = \frac{A + A^T}{2} x + b$$
$$= Ax + b \text{ if } A^T = A$$
$$f(x) = g(Ax + b) \quad \Rightarrow \nabla f = A^T \nabla g(Ax + b)$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$

JACOBIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m \quad x_0 \in \mathbb{R}^n$$
$$J_f(x_0) : \mathbb{R}^{m \times n} \quad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j} f_i(x_0)$$
$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad J_f(x, y) = \begin{pmatrix} \partial x_1 & \partial y_1 \\ \partial x_2 & \partial y_2 \end{pmatrix}$$

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Linearisation:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x_0 \in \mathbb{R}^n$$
$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad g: \mathbb{R}^m \rightarrow \mathbb{R}^k, \quad h: \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$$
$$J_h = J_g(f(x)) \cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M \cdot x \Rightarrow J_f = M$$
$$f(x) = x \quad \Rightarrow J_f = \mathbb{1}$$
$$f(x) = \|x\|^2 \quad \Rightarrow J_f = (2x)^T$$

HYPER-SURFACES

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad g: \mathbb{R}^m \rightarrow \mathbb{R}, \quad h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$$
$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f) |_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad g: \mathbb{R}^n \rightarrow \mathbb{R}$$
$$\nabla(f + g) = \nabla f + \nabla g$$
$$\nabla(f - g) = \nabla f - \nabla g$$
$$\nabla(f \cdot g) = g \nabla f + f \nabla g$$
$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$$f: D \rightarrow M$$
$$g: N \rightarrow R$$
$$h: D \rightarrow R$$
$$= g \circ f$$

GEOMETRY OF HYPER-SURFACES

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

Tangent space: $\alpha, \beta \in \mathbb{R}, T =$

$$\left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$
$$\left\{ f(x_0, y_0) + \alpha \left(\frac{\partial}{\partial x} f(x_0, y_0) \right) + \beta \left(\frac{\partial}{\partial y} f(x_0, y_0) \right) \right\}$$

Graph of a linearisation:

$$l(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) = T_{p_0}$$

SECOND DERIVATIVE

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} f(x, y) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$$

HESSIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n \times n}$$
$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$
$$\partial_i \partial_j f = \partial_j \partial_i f$$

Common Hessians:

$$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$$

$$f(x) = g(Ax + b) \quad \Rightarrow H_f = A^T H_g(Ax + b) A$$
$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$

EXTREMAL POINTS

Quadratic form:

$$M \in \mathbb{R}^{n \times n} \quad q_M(v) = v^T M v \quad v \in \mathbb{R}^n$$
$$H = \frac{1}{2} (M + M^T) \quad \forall v \in \mathbb{R}^n. \quad (q_M(v) = q_H(v))$$

$$v^T H v > 0 \Rightarrow c_0 \text{ is a local minimum, } H \text{ is positive definite}$$
$$v^T H v \geq 0 \Rightarrow c_0 \text{ is a local minimum or non-extremal, } H \text{ is positive semi-definite}$$
$$v^T H v < 0 \Rightarrow c_0 \text{ is a local maximum, } H \text{ is negative definite}$$
$$v^T H v \leq 0 \Rightarrow c_0 \text{ is a local maximum or non-extremal, } H \text{ is negative semi-definite}$$
$$v^T H v \neq 0 \Rightarrow c_0 \text{ is a saddle point, } H \text{ is indefinite}$$

TODO:

$$\partial_j(u_i v_i) = (\partial_j u_i) v_i + u_i (\partial_j v_i)$$

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$
$$= (v_1, v_2, v_3) \begin{pmatrix} v_1 h_{11} + v_2 h_{12} + v_3 h_{13} \\ v_1 h_{21} + v_2 h_{22} + v_3 h_{23} \\ v_1 h_{31} + v_2 h_{32} + v_3 h_{33} \end{pmatrix}$$
$$= v_1^2 h_{11} + v_1 v_2 h_{12} + v_1 v_3 h_{13} + v_1 v_2 h_{21} + v_2^2 h_{22} + v_2 v_3 h_{23} + v_1 v_3 h_{31} + v_2 v_3 h_{32} + v_3^2 h_{33}$$
$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

Technique of completing squares:

$$\frac{x^2 - 4xy + 4}{\sqrt{\quad} \div 2x}$$
$$= (x - 2y)^2 - (-2y)^2 + 4$$
$$= (x - 2y)^2 - 4y^2 + 4$$

GRADIENT DESCENT / NEWTON'S METHOD

$$GD \quad x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$$
$$N \quad x_{i+1} = x_i - \gamma \cdot (H_f(x_i))^{-1} \cdot \nabla f(x_i)$$
$$\gamma = \text{step size}$$

termination criterion $\epsilon > |x_{i+1} - x_i|$

PROBABILITY THEORY

Term	Definition
$\sim$	Distributed
$\mathcal{N}(\mu, \sigma)$	Normal distribution
$\text{unif}(a, b)$	Uniform distribution
$\mathbb{P}(E)$	Probability measure
$l(m, q)$	Likelihood function
$\ln(l(m, q))$	Log-likelihood function
CDF	Cumulative Distribution Function F
PDF	Probability Density Function f

TODO:

formulas

TODO:

$$\text{density} = f(x_i \mid \mu, \sigma)$$
$$\text{likelihood} = \prod_{i=1}^n f(x_i \mid \mu, \sigma)$$
$$\text{log-likelihood} = \dots$$

KOLMOGOROV AXIOMS

$$\mathbb{P}(\Omega) = 1$$
$$\mathbb{P}(\emptyset) = 0$$

$\forall A, B \subset \Omega, \quad A \cap B = \emptyset. \quad (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$
 $\forall A, B \subset \Omega. \quad (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$

STATISTICALLY INDEPENDENT
 $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ are **statistically independent**, if the probability density f_V of the random variable

$$V : \begin{cases} \Omega \rightarrow \mathbb{R}^2 \\ \omega \mapsto \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} \end{cases}$$

satisfies

$$f_V(x, y) = f_X(x) \cdot f_Y(y)$$

DISCRETE PROBABILITY DISTRIBUTION
Let $\Omega = \{e_1, e_2, \dots, e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, \dots, n\}, \quad E \subset \Omega$$

$$\mathbb{P}(E) = \sum_{e \in E} p(e)$$

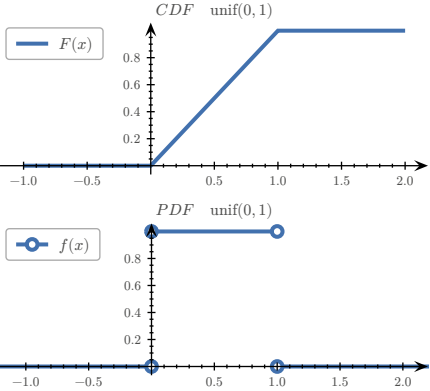
$$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$$

RULES

$$\forall e \in \Omega. \quad (0 \leq p(e) \leq 1)$$

$$\begin{aligned} \sum_{e \in \Omega} p(e) &= 1 & \int_{-\infty}^{\infty} f(t) \, dt &= 1 \\ \lim_{x \rightarrow -\infty} F(x) &= 0 & \lim_{x \rightarrow \infty} F(x) &= 1 \\ 0 \leq F(x) &\leq 1 & f(t) &\geq 0 \text{ for } t \in \mathbb{R} \end{aligned}$$

$F(x)$ is monotonically increasing & continuous from the right



UNIVARIATE UNIFORM DISTRIBUTION

$$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a; b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) \cdot \frac{1}{b-a} = F'_X$$

$$\begin{aligned} F_X(x) &= \mathbb{P}((-\infty; x] \cap \Omega) = \mathbb{1}_{[a;b)}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b;\infty)}(x) \\ &= \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) \, dt \end{aligned}$$

UNIVARIATE NORMAL DISTRIBUTION

TODO:

Standard normal distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

General normal distribution:

$$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$$

Univariate normal distribution:

$$x \sim \mathcal{N}(\mu, \sigma)$$

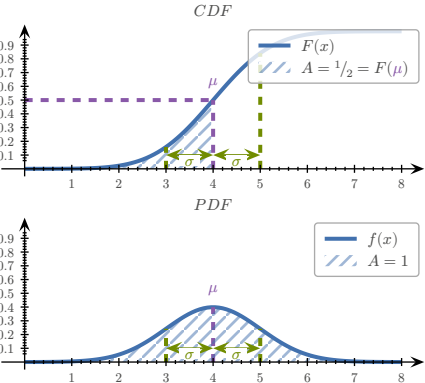
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$\mu \approx \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\sigma^2 \approx \text{var} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

Error function:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, dt$$



mean value (μ), standard deviation (σ)
MULTIVARIATE NORMAL DISTRIBUTION
TODO:

$$\Sigma = \text{covariance matrix}, \quad V : \Omega \rightarrow \mathbb{R}^n$$

$$V \sim \mathcal{N}(\mu, \Sigma)$$

$$f_V(v) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \cdot e^{q_{\Sigma^{-1}}(v-\mu)}$$

$$q_{\Sigma^{-1}}(v-\mu) = -\frac{1}{2}(v-\mu)^T \Sigma^{-1} (v-\mu)$$

Sample mean:

$$\mu \approx \bar{v} = \frac{1}{N} \sum_{i=1}^N v_i$$

Sample covariance matrix:

$$\Sigma \approx Q = \frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})(v_i - \bar{v})^T$$

EXAMPLES

TODO:

- calculating jacobian
- calculate stationary points
- calculate hessian
- calculate probability density function
- If X and Y are statistically independent
- gradient descent
- negative log-likelihood
- find formula for likelihood function
- check if function is increasing ($f' > 0$) $\rightarrow$ Aufgabe 123/124
- distribution to density and vice-versa $X = g(Y) \wedge g \text{ increasing} \Rightarrow f_Y(y) = f_X(g(y)) \cdot g'(y)$
- FS2024 7.b)
- FS2024 5.b)
- FS2024 6.a)b)

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

WORKING WITH INDICES

$$x, y \in \mathbb{R}^n, \quad f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(x) = (x-y)^T (3x-y) = \sum_{i=1}^n (x_i-y_i)(3x_i-y_i)$$

$$\partial x_i f(x) = (3x_i-y_i) + 3(x_i-y_i)$$

$$f(x) = \ln(x_1 \cdot x_2 \cdot \dots \cdot x_n) - |x|^2 = \sum_{i=1}^n (\ln(x_i) - x_i^2)$$

$$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$$